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Farmers and Technology of Future Danish Agrilogistics

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Abstract

This PhD project explores the role of technology in Danish future farming with Research through Design methods to nuance existing future farming imaginaries and elicit new ones. The contemporary agricultural production in Denmark has become deadlocked and faces a "crisis of imagination" where dominant paradigms are hard to challenge, leading to future farming imaginaries that either rely on high-tech solutions or their opposite of back-to-basics rural bliss. This research is structured after the Ethnographic Experiential Futures framework and uses participatory design (games) and co-speculation to make room for alternative imaginaries and understand those of the status-quo.

CCS Concepts

• **Human-centered computing → Participatory design**.

Keywords

Agriculture, Agrilogistics, Farming, Food-systems, Sustainability, Design games

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1 Introduction

Aiming to expand notions of what tomorrow's food production could be like, my PhD research inquires into the role of technology in Danish future farming and what the future farmers hope and expect. Today, Danish farming is a highly rational and efficient endeavor; in other words, it's a sector deadlocked by its own optimization regime facing a "crisis of imagination" [10], making the dominant paradigms hard to challenge. Images of future farming tend to feature either high-tech, automated, digital, and internet-connected cultivation, or the polar opposite, back-to-basics, rural bliss. The disparity between these visions leaves little room for the coming generations of farmers to make the farms of tomorrow into their own. To seek out alternatives, to the currently prevailing future farming imaginaries, my work uses co-speculation, participatory design (PD) design games played with students at agricultural schools, fieldwork, and qualitative interviews with early-career smallholder farmers and other domain experts. The project is structured after the Ethnographic Experimental Futures (EXF) framework [40] to help make sense of the alternative images produced, how they are perceived by the publics around the issues in question, as well as probing what can further develop them.

In Denmark, where this work takes place, there is a long and proud history of farming, and farmers once made the countryside buzz from cultivating the land in individual small holds. Like most places, Danish farms have since been optimized, enlarged, or merged with others, and less than 6,300 full-time farms remain of what was almost 200,000 in the 1960s [38, 39]. The agricultural sector occupies nearly 60% of the country's area, mainly for producing more than 200 million animals [31]. This highly specialized production means that the domestic production cannot provide Danes with the diet recommended by the state [45] and that it contributes about a fourth of the total national greenhouse gas emissions [31]. Nitrogen leaching, caused by excessive spreading of slurry, also means that most inner waters suffer from oxygen depletion, the affected areas larger than Funen and Zealand when combined [22]. Timothy Morton has described this industrial mode of contemporary farming as *agrilogistics*, which emphasizes the fallout of contemporary food systems, the paradox of relying on the Earth while exploiting it, and bending natural eco-systems to fit man-made ones instead [32]. As the fallout of agrilogistics in Denmark has become more apparent, it is now widely acknowledged that the status quo is not viable. That, in turn, gave rise to recently concluded "tripartite negotiations"–named after the labor market structure–between agriculture, industry, and environmental advocates and municipalities. The agreement reached in these tripartite negotiations was expectedly disappointing to many, resulting primarily in voluntary agreements, ill-defined goals, and carbon tax measures effectively lower than previously recommended by a government-appointed committee [17]. Den Grønne Ungdomsbevægelse (DGUB), the Green Youth Movement, denounced political decision-making like the tripartite, writing in 2024 "we are in the very middle of a 'forestillingskrise' (crisis of imagination), and it takes great new dreams for us to begin to faintly see the contours of a future for the agricultural sector within the planetary, economically, and socially sustainable boundaries" [10]. While Danes now mostly agree that there are issues with domestic agrilogistics, *how* to transition to better alternatives, and what those could be like, is still highly contested.

Both within and outside of academia, technological innovation is a common answer to how issues like the above should be addressed, and technology has always been integral to farming. Its development has already consistently increased farmers' harvest yields and lessened the need for manual labor. As ICTs became cheaper and

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more readily available, they too fused with farming at scale. This hybrid, *Digital Agriculture* (DA), now seems inevitable whenever future (sustainable) farming is discussed or envisioned [7, 20]. Many HCI and interaction design researchers have also worked with the domain, as shown by Doggett et al.'s in a 2023 literature review [14]. Here, they also found that projects about ICTs for Development related to farming tend to focus on the Global South, using a top-down approach framed around a "need to address poverty at the individual and national level as well as to respond to the demands of a rising population". By contrast, the projects situated in the Global North focus mostly on urban environments, often framed around "urban population growth, the need for climate action, and the need to collectively imagine what the future of food systems should look like in urban places". The aforementioned agrilogistics were brought into HCI by Biggs et al. [3], who used Morton's framework [32] to analyze smallholder farmers' practices to offer insights in support of responsible designs of sustainable agriculture computing systems. Design researchers have also been giving food systems more attention, inquiring into the particulars of food, including human/soil relations [30], speculations into multi-species care [16], foraging [13], digestible data [43], and communal dining [37]. These projects' approach to food systems, via food as a design object in itself or through social engagement, seems a natural progression for object- and community-oriented design research disciplines. But like the approaches often chosen in HCI, these also leave large-scale food production underexplored.

Besides the limited attention to the transitioning of dominant, large scale food production, HCI researchers' assumptions about agricultural technology tend to be based on their own experiences which are often in stark contrast to the realities of rural infrastructure. These perspectives can also discredit farmers' historical role as producers of infrastructure and contemporary role as tinkerers and adapters of technology to suit their own needs [35]. Hardy et al. also pointed to urban spaces' domination of HCI research, arguing it leads to rural residents being overlooked but that attention to rural communities can "broaden perspectives on who we design for and what practices are worth designing for" [23]. Responding to the underrepresentation of farmers, my project engages main course students at farming schools, a group that finds itself in a space dominated by larger forces. Despite that most farming students are not (yet) independent, main course students already have farming experience and insight from internships that follow the basic course. More importantly, they are invested in the future of the Danish farming.

My research can hopefully uncover some of that currently left unturned but is also driven by personal interest. It grew after I finished a B.A. in architecture at the Royal Danish Academy in 2017 and spent most of the next year learning about farming at a Danish farming school and working at farms in Japan and Tanzania. This did not make me a farmer but humbled, and left me with deeply felt respect for farmers–big or small, conventional or organic–from having my ignorance revealed of farmers' working and living conditions. I also became eager to learn more about this sector, relevant to all who need to eat, facing fundamental shifts in the times to come. These experiences have also influenced the structure and approach of the PhD project, mainly by drawing on interdisciplinarity and farmers' expertise.

2 Research Questions

Many researchers are already producing perspectives and innovation bringing us closer to sustainable food production, but projects that directly engage with the transition of the most dominant production forms remain sparse. To challenge and expand notions of what can follow the slow-motion crisis that is Danish agrilogistics, this project asks:

- What is the role of technology in the green transition of Danish food production?
- How do contemporary agrilogistics produce currently dominant futures for farming?
- What "preferable" visions offer viable alternatives to existing agrilogistic practices?
- How can research through design manifest responsible imaginaries to enable dialogue on preferable farming futures?

3 Theoretical Approaches

By its nature, design is invested in the future, with many design tools being rooted in seeking out and articulating what hasn't yet happened. Sketches, storyboards, and especially scenarios, capture and convey the future as it is to be built, but the future has been brought into design more theoretically as well. As a research discipline, futures and futuring give a language and framework to describe and generate different futures. Stuart Candy described Dator's four "generic futures" of continue, collapse, discipline, and transform as broadly representing the stories that we tell about the future, with the categories offering ways to think concretely about what ideas of the future do, as well as to project into the future in more varied ways than as "utopia" or "dystopia" [8, 11]. Ideas from foresight were imported into design via Dunne and Raby's book *Speculative Everything* [15] and have become the basis of a wide range of methods that critique and expand on how we imagine futures and, critically, who gets to do that imagining [24].

The use of speculative design in HCI research projects for sustainability was examined by Soden et al. [36], who found it has gained traction in the subgenres Sustainable HCI (SHCI) and Sustainable Interaction Design (SID). Despite stated benefits to this, like questioning values and ethics embedded in current practices and promoting public discourse on climate change, sustainability, and the future, Soden et al.'s analysis left them discouraged. Speculative design was most often used to present positive visions of the future or problem framing but rarely to inspire new narratives that could lead to new ways of living. In agreement with critics calling for meaningful reorientations of speculative design, needed for it to realize its critical potential, Soden et al. argue for uses "grounded in a positive project that has some notion of how society transitions to sustainability and a sense of urgency about this process". DiSalvo has relatedly discussed the use of "spectacle" and "tropes" in speculative design projects [12], writing that "[n]ot all speculative design must be political, but to not address politics in social contexts where they are usually present is a striking omission". Aligned with Soden et al. [36], he concluded that, "[t]oo often, speculative design is spectacle alone (…) lacking the content and grounding to make productive critical statements or spark public debate". Design futuring approaches, as understood in this project, do not offer solutions or targets but rather aim to open up critique,

debate, and questioning of the future to seek alternative futures, as well as address and potentially change the present [28]. Like so, the future is an object of discussion meant to both cohere current political concerns and leave the future open to be imagined in new, less constrained, speculative ways [26]. While the future is not bound by the present, it does come from it, so developing new narratives and means of futuring could help engage and allay a crisis of imagination in times of crises.

My project deals with sustainability, but, as informed by perspectives like the above, it does not assume to know what kind of sustainability will be best for future farmers. Yet, HCI scholars have already explored what sustainability can entail in contexts of technological development, HCI, and interaction design. Baumer and Silberman, for example, argued for improving the conduct of these domains and urged researchers to interrogate the appropriateness of IT in a given situation [2], and Blevis suggested principles for designing sustainably, like "linking invention & disposal" and "promoting renewal & reuse" [4]. To steer clear of assumptions on what's best for future farmers in this project, SHCI and SID literature is not regarded as input for developing "better" or more sustainable solutions but rather for understanding the currently dominant future farming imaginaries and making sense of those produced in the project. Practically, it offers guidance for doing the research while striving to be thoughtful and sustainable.

4 Methods

The project is planned as a Research through Design (RtD) process [19] and structured in accordance with Candy and Kornet's Ethnographic Experiential Futures (EXF) framework [40]. The work relies on ethnographic engagement with farming students, leveraging participatory design (PD) games [5, 6, 42] as a form of inquiry, focusing on documentation [1] of the PD engagements, broader process, and iteration of the materials. EXF draws and builds on various ways of thinking on the future to offer a cyclical five-step structure [9] for futuring in RtD research like mine.

Step 1) *Mapping* involves inquiry into people's existing images of the future. In this project, it is done primarily through interviews, fieldwork, desk-research, and with a purpose-built design game [34] to be played with farm students and farmers. This game reinterprets the Landscape Game [5, 21] but, drawing on Ingold's notion of *landscape* as a relational, temporal space for *dwelling* [25], it takes "landscape" more literally than its exemplar. The goal of this project's landscape game is to create opportunities for future farmers to situate their desired farms within broader social, environmental, and technical contexts while exploring how they and their surrounding landscapes might evolve over time. This work will take place over the Summer and Fall of 2025.

Step 2) *Multiply* is about generating alternative future images. The data and images from the mapping phase are coded, analyzed, and iterated on, and will be done partly in collaboration with Will Odom's Homeware Lab at Simon Fraser University during a prospective research stay. For this work, approaches like *fabulation* [41] or *infrastructural speculations* [44] can help produce materials like workbooks [18] to outline a design space and formulate design briefs that can be used in the following step.

Step 3) *Mediate* concerns translating the alternative future images into experiential manifestations and is planned to happen in a workshop with design or architectural students at a Danish arts-based education. Broadly, these students will be tasked with giving form to what could be the farming students' future farms, using prototypes [27, 29, 33], like models, collages, or architectural drawings. Photo-realistic renderings are not the sought-after output of the workshop; it should rather produce conceptual depictions to suggest and open up notions of what future farms could be like, and not serve as fully fledged answers.

Step 4) *Mount* stages the new (futures) representations for participants to experience. This will put materials of the latter workshop on display in a form of exhibition that can serve as a basis for the farming students, and other interested parties, to react, evaluate, and discuss them. Ideally, this will also serve as a space for starting a conversation between the two student groups. For contributing students who cannot visit the exhibition in person, an online or scaled-down version is to be made available.

Step 5) *Map II* involves recording the responses to the new representations, or experiential scenarios. Depending on venue, media, and practicalities, the second mapping can happen both in situ and over time. For more qualitative evaluation, participants of the earlier steps, especially the students at farm-schools, are invited for feedback and discussion sessions.

5 Conclusion

This document outlines my PhD research project that investigates the role of technology in future farming through participatory engagements with students at Danish agricultural schools. The research aims to produce new future farming imaginaries, cultivate productive discourse on present agricultural practices, and to offer methodological contributions on using the Ethnographic Experiential Futures framework for producing the new future farming imaginaries with participatory design games and co-creation of materials.

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